
Real Data Lies: Unveiling and Closing the Quality Shortcut in Generalizable AI-Generated Video Detection

Ziyuan Fang¹ Tianyi Wei^{✉ 2} Guanjie Wang¹ Weiming Zhang¹ Nenghai Yu¹ Wenbo Zhou^{✉ 1}

Abstract

Recent advances in video generation have enabled highly realistic synthetic content, raising concerns about the integrity of digital media and motivating the development of benchmarks and detection methods for generated videos. Prior works have largely prioritized bolstering model generalization against unseen generators. However, we uncover a neglected factor: the quality distribution of real videos plays a pivotal role. Current training protocols suffer from a clear quality bias between real and fake data, prone to shortcut learning. Compounded by testing on similar real data distributions, this creates an illusion of generalization. In reality, these models fail to generalize when exposed to real data with significantly different quality profiles. To address this, we propose training with quality-matched real and fake data to mitigate bias. Building on this, we introduce a data expansion strategy that broadens the training set to comprehensively cover the full quality spectrum. This approach enables the model to learn quality-agnostic features for detection, thereby achieving generalization across real data of varying qualities and enhancing real-world applicability. Extensive experiments demonstrate that our method scales well across diverse backbones, consistently enhancing the generalization capability of existing models. Code: [GitHub](#)

1. Introduction

Recently, generative models have achieved unprecedented breakthroughs ([Google DeepMind, 2025](#); [OpenAI, 2025](#)),

¹School of Cyber Science and Technology, University of Science and Technology of China, Hefei, Anhui 230026, China.

²College of Computing and Data Science, Nanyang Technological University, Singapore, 639798, Singapore. Correspondence to: Tianyi Wei <besttianyiwei@gmail.com>, Wenbo Zhou <welbeckz@ustc.edu.cn>.

expanding rapidly from static images to photorealistic videos. Notably, the release of Sora ([Brooks et al., 2024](#)) marked a pivotal shift, catalyzing a surge in high-fidelity video synthesis. However, the widespread proliferation of such hyper-realistic content undermines the authenticity of digital media and escalates concerns regarding malicious misuse.

To address these challenges, a number of benchmarks and detection algorithms have been introduced by the research community. Given the rapid iteration and diverse nature of generative models, exhaustively covering every synthesis method is impractical. Consequently, existing approaches ([Chen et al., 2024a](#); [Zhang et al., 2025b](#); [Zheng et al., 2025](#); [Li et al., 2025b](#)) primarily aim to achieve robust generalization against unseen models, a priority also reflected in current benchmark designs ([Chen et al., 2024a](#); [Ni et al., 2026](#); [Wen et al., 2025b;a](#)).

Despite these efforts, our empirical analysis uncovers a perplexing inconsistency. Specifically, our experiments reveal a critical phenomenon: various detectors trained on GenVideo ([Chen et al., 2024a](#))—the most widely adopted yardstick for benchmarking generalization to unseen generative models—exhibit drastically different performance behaviors when evaluated on different real datasets. Notably, we observe that models suffer from severe performance degradation on certain real datasets, while maintaining reasonable effectiveness on others. Crucially, these variations follow a consistent, model-agnostic pattern, resembling a clustering effect where specific datasets yield similar performance traits. This observation indicates that models trained on GenVideo have universally overfitted to the real data within the training set. Furthermore, the existing testing protocol of GenVideo fails to capture this significant deficiency.

To pinpoint the root cause of this overfitting, we conducted an in-depth analysis of the real datasets. We employed a video quality assessment (VQA) model, DOVER ([Wu et al., 2023](#)), to evaluate and statistically analyze the videos in GenVideo. The results reveal a clear trend: real datasets where models perform well exhibit significantly lower quality than that of generated data. Conversely, model performance degrades severely on real data with quality comparable to or exceeding the average level of generated samples. More-

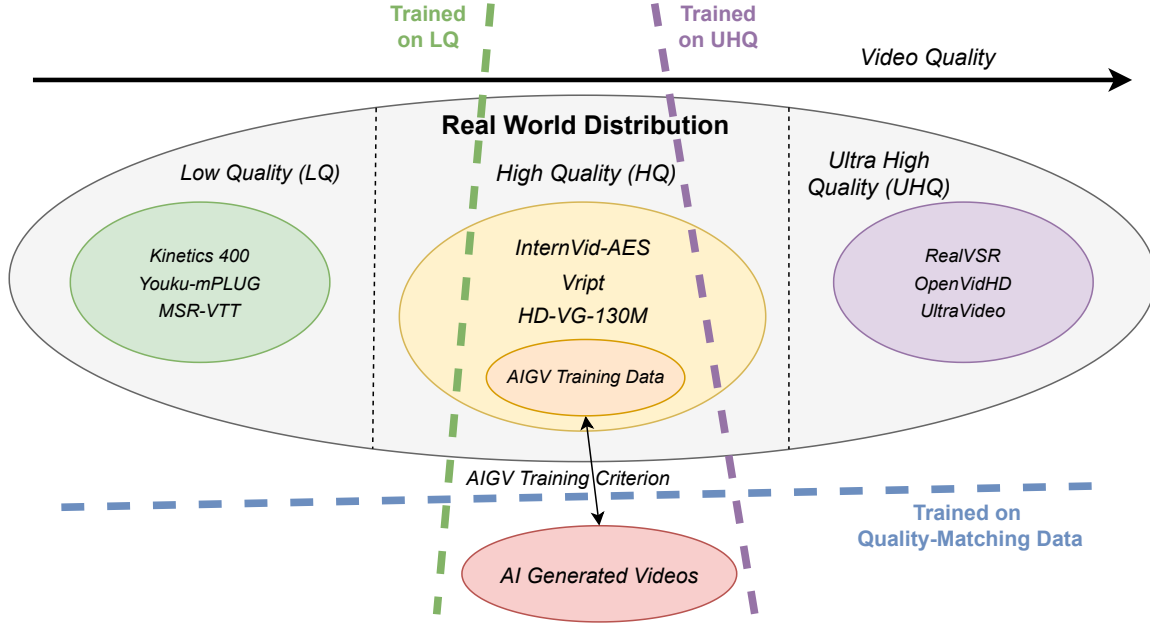


Figure 1. Illustration of the impact of data quality distribution. We discover and demonstrate that when there are significant statistical discrepancies in quality between real and generated data, models tend to exploit these quality cues for discrimination, resulting in the formation of flawed decision boundaries.

over, the average quality of the real data used in the GenVideo training set closely aligns with that of the specific real datasets where the model performs well. Based on these findings, we posit that the substantial quality gap—where *training real videos are of significantly lower quality than generated ones*—causes models to rely on quality cues for discrimination. Consequently, this learned decision boundary collapses when the model encounters real videos with quality comparable to or higher than the generated samples.

To further verify this proposition and explore methods to mitigate this overfitting, we performed extensive experiments on a diverse set of real datasets characterized by distinct quality levels. Our experimental findings lead to the conclusion that training on real data with quality either substantially lower or higher than generated samples causes the model to exploit quality differences for discrimination, establishing a flawed decision boundary, as depicted in Fig. 1. In contrast, we demonstrate that training with quality-matched real and fake data effectively mitigates this bias, avoiding quality-specific overfitting.

Guided by this conclusion, we propose a training strategy grounded in the principle of quality matching to align real and fake data distributions, thereby severing quality-based shortcuts. Building upon this alignment, we introduce a quality-aware data expansion strategy that augments the training set to comprehensively encompass the *full quality spectrum*. This approach empowers the model to learn quality-agnostic features, ensuring robust generalization

across real data of varying qualities and enhancing real-world applicability. Extensive experiments demonstrate that despite its simplicity, our method scales effectively across diverse backbones, significantly bolstering the generalization capabilities of existing models.

Complementing our proposed training paradigm, we refine the GenVideo evaluation protocol by incorporating real data exhibiting diverse quality levels. This initiative aims to establish a more rigorous framework, enabling a comprehensive and accurate assessment of detector generalization and real-world applicability. Crucially, this revision rectifies a systemic deficiency pervasive across existing benchmarks (e.g., GenVideo, GenVidBench, GenBuster-200K, and GenBuster++), where the real data utilized for both training and evaluation invariably exhibits homogeneous quality profiles confined to a narrow slice of the full quality spectrum. Our revised protocol effectively bridges this critical gap.

Our contributions are summarized as follows:

- We identify and experimentally validate a critical shortcut learning phenomenon in AI-generated video detection, stemming from the quality distribution discrepancy between real and synthesized videos. Furthermore, we elucidate the underlying mechanism driving this behavior.
- We propose a simple yet effective training paradigm to mitigate this bias. By enforcing quality matching to align distributions and introducing a quality-aware

expansion strategy to encompass the full quality spectrum, our method compels the model to learn quality-agnostic features, successfully blocking the quality-based shortcut. Extensive experiments demonstrate that our approach exhibits remarkable scalability across diverse backbones, consistently bolstering their generalization capabilities.

- We expose the design deficiencies within existing benchmarks on both the training and evaluation sides. Consequently, we introduce a revised protocol that provides a more comprehensive and accurate assessment of a detector’s true generalization capabilities.

2. Related Works

2.1. AI Video Generation

Recently, video generation has witnessed a breakthrough. Representative closed-source models (e.g., Sora (OpenAI, 2025), Kling (Kuaishou KlingAI, 2025)) and open-source frameworks (e.g., HunyuanVideo (Kong et al., 2024), Wan (Wan et al., 2025)), collectively showcase remarkable improvements in diversity, quality, and iteration speed. Furthermore, as the usage threshold lowers, these powerful tools are becoming pervasive. Consequently, the high realism and rapid diffusion of generated videos are creating unprecedented threats to the trustworthiness of digital media. It is worth noting that, aiming for high-fidelity synthesis, almost all models employ rigorous data filtering and curation mechanisms based on quality metrics. Then the models are explicitly optimized to mimic the high-quality distribution of this selected data. This standard paradigm, encompassing both data filtering and training objectives, is widely documented in the technical reports of mainstream video generation models (Polyak et al., 2024; Kong et al., 2024; Wan et al., 2025; Wu et al., 2025). Consequently, this inherent trait results in the quality distribution of generated videos being concentrated in a specific, generally high, band of the quality spectrum.

2.2. AI-Generated Image Detection

AIGV detection is closely related to AI-generated image detection, as videos consist of frames carrying generative artifacts. CNNSpot (Wang et al., 2020) first achieves favorable generalization to GAN-generated images by training a standard image classifier. To address emerging diffusion models, DIRE (Wang et al., 2023d) proposes detecting diffusion-generated images based on diffusion reconstruction error. UnivFD (Ojha et al., 2023) improves detection generalization from GANs to diffusion models by leveraging a feature space not explicitly trained to distinguish real from fake images. NPR (Tan et al., 2024) introduces neighboring pixel relationships to capture structural artifacts stemming from

up-sampling operations. SAFE (Li et al., 2025a) enhances detector generalization via three simple image transformations. AIDE (Yan et al., 2025) adopts a mixture-of-experts architecture to simultaneously extract visual artifacts and noise patterns. Notably, DDA (Chen et al., 2025) employs 2D frequency-domain processing and image-based VAE reconstruction to align real and fake data, which bears some similarity to our approach. However, modern video generators increasingly adopt video-level VAEs and temporal modules (Polyak et al., 2024; Kong et al., 2024; Wan et al., 2025; Wu et al., 2025), and image-level reconstruction may compromise temporal consistency in videos. By contrast, our selection-based approach achieves alignment through intelligent sampling while preserving the authenticity of both real and generated video content.¹

2.3. AI-Generated Video Detection

GenVideo (Chen et al., 2024a) currently serves as the most recognized benchmark for AIGV detection with its diverse collection, alongside the proposed DeMamba method (Gu & Dao, 2024) for capturing local inconsistencies. To address evolving generation capabilities, recent benchmarks have introduced more challenging settings: GenVidBench (Ni et al., 2026) emphasizes Cross Source and Cross Generator evaluation with rich semantic labels, while GenBuster-200K (Wen et al., 2025b) and GenBuster++ (Wen et al., 2025a) focus on large-scale commercial model outputs and rigorously curated cross-modal samples, respectively. In terms of detection methodologies, NSG-VD (Zhang et al., 2025b) and D3 (Zheng et al., 2025) have recently introduced physics-based perspectives to the field. By leveraging probability flow conservation principles and second-order dynamical analysis under Newtonian mechanics respectively, these methods detect AI-generated videos by capturing their violations of physical motion laws. Furthermore, BusterX (Wen et al., 2025b), BusterX++ (Wen et al., 2025a), and Skyra (Li et al., 2025b) propose leveraging Multimodal Large Language Models (MLLMs) to achieve explainable forgery detection. In contrast to these works, we unveil an empirical regularity governing model generalization on real data, aiming to provide theoretical underpinnings for future benchmark construction and method design.

3. Impact of Data Quality Distribution on Model Generalization

Overview. In this section, we investigate the impact of data quality distribution on model generalization. Initially, we empirically reveal that detectors trained on GenVideo—currently the most influential AIGV detection benchmark—

¹We experimentally compare with DDA and provide analysis in Appendix B.4, demonstrating that DDA does not directly transfer to AIGV detection.

exhibit severe overfitting to the real data within the training set. Subsequently, through comprehensive video quality statistical analysis and training experiments on a series of real datasets with distinct quality levels, we pinpoint the root cause of this phenomenon: *training on real data with a quality distribution significantly divergent from that of generated samples induces the model to exploit quality discrepancies as a discriminative shortcut.*

3.1. Experimental Setup

Dataset and Protocol. We base our experimental setup on GenVideo (Chen et al., 2024a), adapting the one-to-many protocol (i.e., training on a single generator and generalizing to unseen ones) from DeMamba (Chen et al., 2024a) and NSG-VD (Zhang et al., 2025b) into a modified several-to-many protocol (i.e., training on one generator, validating on a distinct second generator, and generalizing to unseen models). The specific data partition is detailed as follows:

- Training Set: Consists of 10,000 Pika-generated (Pika, 2025) videos and 10,000 real videos.
- Validation Set: Includes 1,000 SEINE-generated (Chen et al., 2023b) videos and 1,000 real videos.
- Testing Set: Comprises 1,000 real videos and samples from 9 generative models in the GenVideo test suite.²

This protocol simulates real-world scenarios where access is restricted to a limited subset of generative models. Crucially, we do not specify a fixed real dataset for the training set in this general protocol. Instead, the source of real training data acts as a controlled variable in subsequent experiments to investigate the impact of quality distribution. Regarding real data, to simulate the high variance of video quality in the wild, we curate a diverse collection of real video sources covering the full quality spectrum. Specifically, we incorporate the real video collection curated in GenVideo (Chen et al., 2024a), including Kinetics-400 (Kay et al., 2017), Youku-mPLUG (Xu et al., 2023), and MSR-VTT (Xu et al., 2016). Furthermore, we integrate Vript (Yang et al., 2024) and HD-VG-130M (Wang et al., 2025a), which are adopted by GenVidBench (Ni et al., 2026), along with OpenVidHD (Nan et al., 2024), which is utilized by GenBuster++ (Wen et al., 2025a). Both GenVidBench and GenBuster++ are emerging AIGV detection benchmarks. To further enrich diversity, we also include InternVid-AES (Wang et al., 2023c), RealVSR (Yang et al., 2021), and UltraVideo (Xue et al., 2025). For more detailed information about these real datasets, please refer to Appendix A.1.

²Among the 10 models in GenVideo, HotShot (Mullan et al., 2023) is excluded as many of its videos have fewer than 16 frames, failing to meet the input requirements of certain detectors.

Evaluation Metrics. To eliminate the influence of classification thresholds on the evaluation results, we adopt the Area Under the ROC Curve (AUC) for a comprehensive assessment of model performance. For image detectors, we average the prediction scores of 8 frames to obtain the final prediction for video-level AUC calculation. Given our focus on the variations in real data, during testing, we pair the 1,000 real videos with videos from each of the 9 generative models individually to calculate the AUC, and then report the average. For generative models with fewer than 1,000 samples, we randomly reduce the number of real videos to maintain a balanced 1:1 real-fake ratio. For video quality assessment, we employ DOVER (Wu et al., 2023) due to its superior alignment with human preferences. Specifically, DOVER outputs both a Technical Score and an Aesthetic Score, which are fused to derive a final Overall Score. In our experiments, we adopt this Overall Score as the holistic measure of video quality, denoted as the DOVER Score.

Detection Models. We employ five image-level models: NPR (Tan et al., 2024), SAFE (Li et al., 2025a), and AIDE (Yan et al., 2025), which are specialized detectors for AI-generated images, alongside DINOv2 (Oquab et al., 2023) and DINOv3 (Siméoni et al., 2025), which serve as general-purpose pre-trained image backbones. Additionally, we utilize four video-level models: TimeSformer (Bertasius et al., 2021) and VideoMAEv2 (Wang et al., 2023b) as generic video processing backbones, while DeMamba (Chen et al., 2024a) and NSG-VD (Zhang et al., 2025b) are specialized architectures explicitly designed for AI-generated video detection. In terms of model specifics, we employ the Large variants of both DINOv2 and DINOv3, and the Base version of VideoMAEv2. TimeSformer is initialized with weights pre-trained on Something-Something-V2 (SSv2) (Goyal et al., 2017). Additionally, we deploy DeMamba-XCLIP, identified as the most effective variant in its original study.

Training Settings. To ensure reproducibility, we consistently set the random seed to 42 for all experiments. For NSG-VD, we adhere strictly to the training configuration described in the original paper. For all other models, we standardize the optimizer to AdamW with a weight decay of 0.1. The learning rate is set to 10^{-5} for image-level backbones (DINOv2 and DINOv3) and 10^{-6} for video-level backbones (TimeSformer, VideoMAEv2, and DeMamba). We set the maximum number of epochs to 30 and employ an early stopping strategy, which terminates training if no performance improvement is observed for 5 consecutive epochs. For evaluation, we select the checkpoint corresponding to the highest AUC on the validation set.

3.2. Impact of Data Quality Distribution

Initial Observation. Adhering to the previously outlined protocol, we utilize Kinetics-400 as the source of real train-

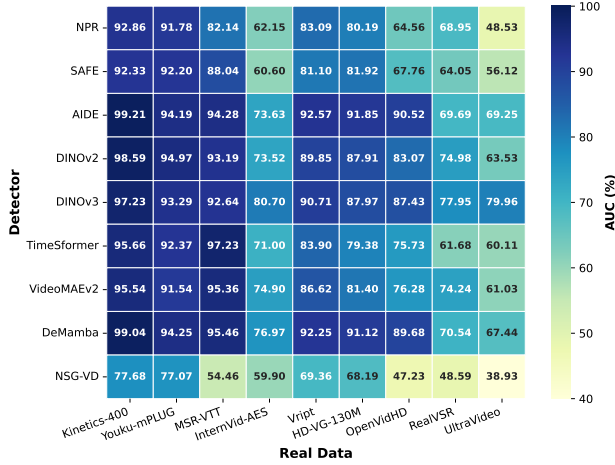


Figure 2. Evaluation of models trained on Kinetics-400 across diverse real datasets, measured by Video-level AUC.³

ing data and evaluate the performance of multiple backbones across a diverse set of real datasets. As shown in Fig. 2, a distinct performance clustering phenomenon emerges. Specifically, while the detectors achieve high accuracy on datasets such as Youku-mPLUG and MSR-VTT, their performance drops significantly on others, most notably on InternVid-AES, RealVSR, and UltraVideo, despite all test sets consisting of genuine videos. This suggests that the observed generalization performance is heavily dependent on the specific distribution of the real data used for evaluation.

Statistical Analysis. Upon visual inspection (representative samples for each dataset are provided in the Appendix A.1), we notice a distinct pattern: real datasets where the model yields high performance typically possess lower visual quality, whereas those leading to performance degradation exhibit significantly higher quality. Furthermore, AIGV data generally exhibits superior visual quality compared to Kinetics-400, Youku-mPLUG, and MSR-VTT—precisely the datasets where the models perform well. To quantify this, we employ the DOVER Score to conduct a comprehensive video quality assessment and statistical analysis across all real and generated datasets. The DOVER scoring results are presented in Fig. 3. It is evident that Kinetics-400, Youku-mPLUG, and MSR-VTT share comparable quality levels, which are substantially lower than the average quality of the AIGV data. In contrast, the average quality of the other real datasets is significantly higher. Specifically, the quality of InternVid-AES closely aligns with the AIGV average, whereas UltraVideo and OpenVidHD markedly surpass it. Accordingly, based on these quality profiles, we stratify the real data into three tiers:

³NSG-VD requires a reference set of real videos. Accordingly, we curated an additional 100 videos from Kinetics-400, ensuring they are strictly disjoint from all experimental splits.

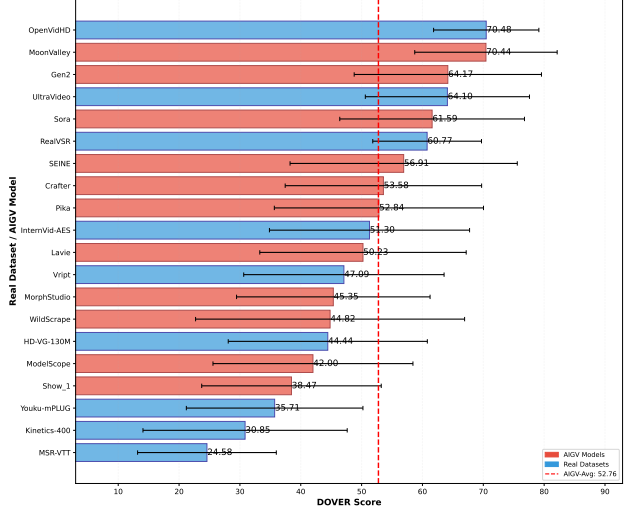


Figure 3. Statistical results of video quality assessment for real datasets and generative models. Here, AIGV-Avg denotes the mean quality of all generative models.

- **Low Quality (LQ):** Includes Kinetics-400, Youku-mPLUG, and MSR-VTT, all sourced from the Gen-Video benchmark.
- **High Quality (HQ):** Consists of InternVid-AES, Vript and HD-VG-130M.
- **Ultra-High Quality (UHQ):** Comprises OpenVidHD, RealVSR and UltraVideo.

Motivated by the aforementioned findings, we formulate the following hypothesis: The systematic quality gap between low-quality training real data (e.g., Kinetics-400) and high-quality generated data (e.g., Pika) traps the model into learning a spurious correlation. Specifically, the detector degenerates into a quality classifier, erroneously using high visual fidelity as a proxy for forgery. Consequently, when the real test data exhibits quality comparable to or higher than the generated data, this inherent defect is exposed, leading to detection failure.

Experimental Verification. To further verify this hypothesis and investigate the relationship between the quality of real training data and model generalization, we conduct experiments by training models separately on three real datasets representing different quality levels: Youku-mPLUG (LQ), InternVid-AES (HQ), and OpenVidHD (UHQ). To ensure experimental consistency, we employ VideoMAEv2 as the unified model for this analysis (We also provide the results using DeMamba as the backbone in the Appendix B.2.).

As the results shown in Fig. 4, the model trained on Youku-mPLUG exhibits a performance pattern similar to that of Kinetics-400: both achieve high accuracy on LQ datasets

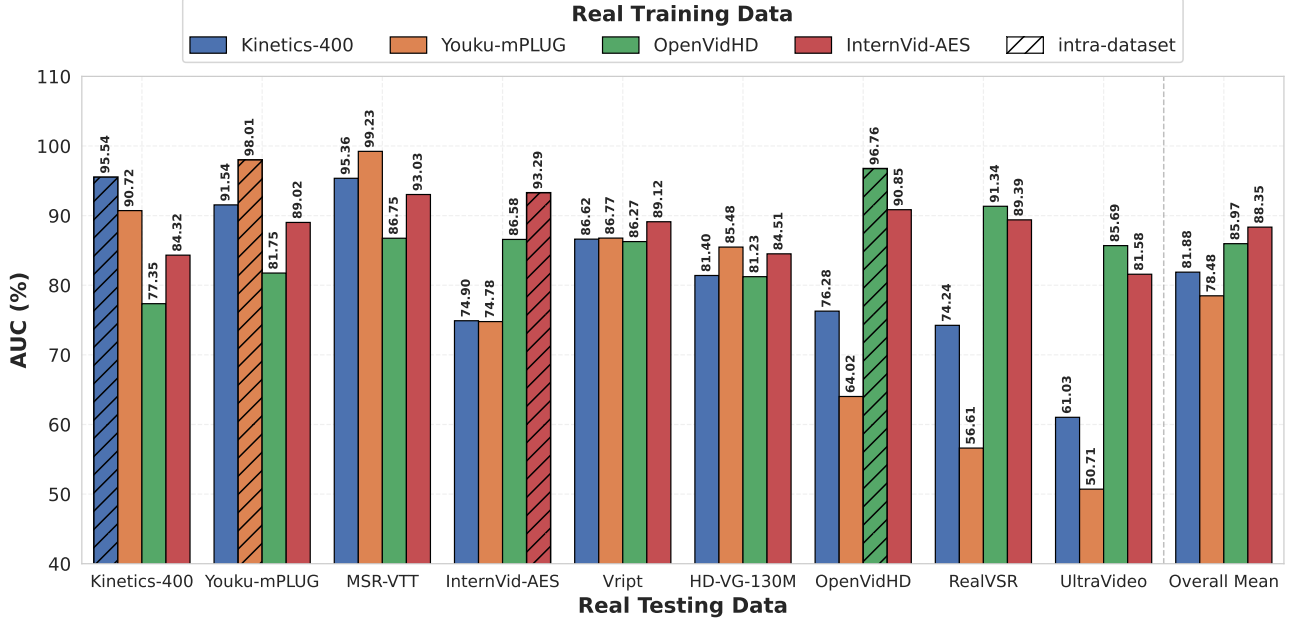


Figure 4. Cross-dataset generalization performance of VideoMAEv2 trained on different real datasets. The hatched bars represent intra-dataset evaluation results.

but suffer significant degradation on HQ and UHQ datasets. Conversely, the model trained on OpenVidHD displays the opposite trend. It generalizes well to RealVSR (another UHQ dataset), yet struggles to generalize effectively to HQ and LQ datasets. More importantly, the model trained on InternVid-AES achieved exceptional generalization results. It maintained high performance standards across all three quality levels and significantly outperformed the other three models in terms of the Overall Mean.

To provide a more intuitive visualization of the quality discrepancies, we also plot the probability density curves of the quality scores for the respective real training datasets and Pika, as shown in Fig. 5. It is evident that the quality distribution of InternVid-AES closely aligns with that of Pika. In contrast, the other datasets exhibit significant distributional discrepancies compared to Pika. Notably, Kinetics-400 and Youku-mPLUG share similar distribution profiles, explaining why they yield such similar results.

In summary, these findings offer solid evidence backing our hypothesis, highlighting the critical role of quality alignment in ensuring reliable detection generalization. Building on this insight, we advocate that the impact of quality distribution be explicitly factored into the design of AIGV detection benchmark. Specifically, training should adhere to the principle of aligning the quality distributions of real and fake data, while simultaneously ensuring comprehensive coverage of the entire quality spectrum, to prevent the model from overfitting to specific quality cues. Correspondingly, evaluation benchmarks should comprehensively cover

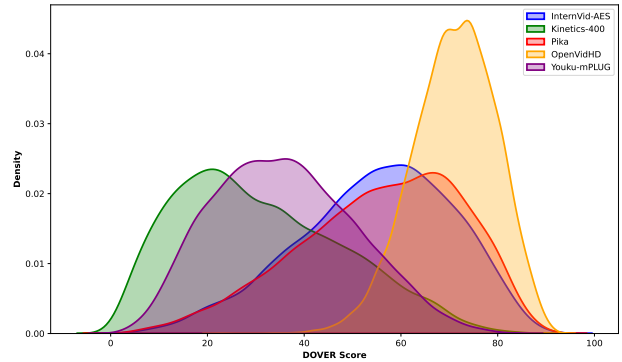


Figure 5. Probability density function (PDF) plots of video quality scores for different real datasets and Pika.

the entire quality spectrum, thereby enabling a more objective and holistic assessment of the model’s generalization capability and real-world usability.

4. Training Data Alignment and Expansion via Quality Matching

Overview. Guided by the principles of quality alignment and coverage, this section introduces a data-driven approach designed to enhance backbone generalization across diverse real data. Additionally, we propose a revised protocol that utilizes a spectrum of real-world datasets for evaluation, addressing the limitations of current benchmarks and filling the critical gap in comprehensive performance assessment. Through extensive experiments, we validate the effective-

Table 1. Validation of quality-aware data alignment and expansion across backbones. Both strategies significantly bolster the generalization performance of existing detectors.

Modality	Model	Training Data	Real Testing Data						Overall Mean
			Youku-mPLUG	MSR-VTT	Vript	HD-VG-130M	RealVSR	UltraVideo	
Image	SAFE	Biased	92.20	88.04	81.10	81.92	64.05	56.12	77.24
		Aligned	85.84	68.56	71.00	76.64	83.00	78.92	77.33 ^{+0.09}
		Expanded	93.22	80.20	82.62	86.83	82.33	85.72	85.15 ^{+7.91}
	AIDE	Biased	94.19	94.28	92.56	91.85	69.69	69.25	85.30
		Aligned	90.76	89.52	93.73	91.42	71.60	85.07	87.02 ^{+1.72}
		Expanded	94.96	92.72	96.18	95.08	89.82	88.76	92.92 ^{+7.62}
	DINOv2	Biased	94.97	93.19	89.85	87.91	74.98	63.53	84.07
		Aligned	94.56	91.88	89.86	85.79	93.29	83.46	89.81 ^{+5.74}
		Expanded	96.37	95.03	95.41	94.26	95.30	92.62	94.83 ^{+10.76}
	DINOv3	Biased	93.29	92.64	90.71	87.97	77.95	79.96	87.09
		Aligned	88.22	85.54	87.89	84.14	85.86	87.90	86.59 ^{+0.50}
		Expanded	95.22	93.04	95.45	94.58	90.49	93.43	93.70 ^{+6.61}
Video	DeMamba	Biased	94.25	95.46	92.25	91.12	70.54	67.44	85.18
		Aligned	95.61	96.88	91.10	86.46	91.80	91.78	92.27 ^{+7.09}
		Expanded	94.26	95.22	95.22	94.92	91.93	94.87	94.40 ^{+9.22}
	TimeSformer	Biased	92.37	97.23	83.9	79.38	61.68	60.11	79.11
		Aligned	82.63	91.50	78.54	73.51	87.48	73.14	81.13 ^{+2.02}
		Expanded	90.60	95.42	88.64	85.57	94.09	85.65	89.99 ^{+10.88}
	VideoMAEv2	Biased	91.54	95.36	86.62	81.40	74.24	61.03	81.70
		Aligned	89.02	93.03	89.12	84.51	89.39	81.58	87.78 ^{+6.08}
		Expanded	93.72	95.21	95.53	91.86	92.48	91.54	93.39 ^{+11.69}

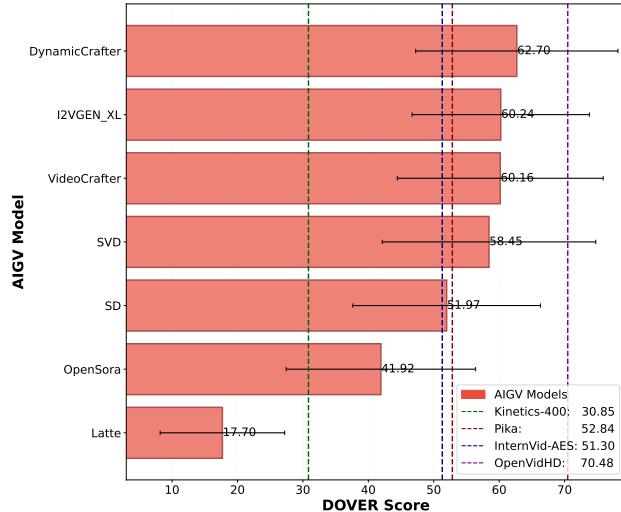


Figure 6. Statistical results of video quality assessment for the remaining generative models within the GenVideo training set.

ness of our proposed data-driven approach, examine its data efficiency and scalability, and verify the advantages of the revised protocol.

4.1. Quality-aware Data Alignment.

Guided by the principles established in the previous section, we first propose replacing the combination of Kinetics-400 and Pika with the InternVid-AES and Pika pair. This substitution ensures quality alignment between real and generated

data while simultaneously providing a certain degree of coverage across various quality bands, as illustrated in Fig. 5.

4.2. Quality-aware Data Expansion

Building upon the foundation of quality matching, we further expand the training data to achieve a more comprehensive and balanced coverage of the entire quality spectrum. Specifically, we select Kinetics-400 and OpenVidHD as the sources for Low Quality (LQ) and Ultra-High Quality (UHQ) real data expansion, respectively. To ensure distributional alignment, we conducted a statistical analysis of the quality distributions of the remaining generative models in the GenVideo training set (see Figure 6). Guided by the principle of minimizing the quality gap, we identified OpenSora (Zheng et al., 2024) as the optimal counterpart for Kinetics-400 and DynamiCrafter (Xing et al., 2024) for OpenVidHD. Finally, we sampled 10,000 videos from each of these sources and merged them with the baseline InternVid-AES and Pika datasets to construct the final expanded training set.

4.3. Protocol Revision and Improvements Across Different Backbones

Protocol Revision. To comprehensively assess the model’s generalization capabilities across diverse real-world distributions, we revise the GenVideo evaluation protocol utilizing six real datasets, balanced across three quality tiers. Specifically, the test sets include Youku-mPLUG and MSR-VTT

Table 2. Ablation study on the proposed quality-aware data expansion. LQ denotes the data pair of Kinetics-400 and OpenSora, while UHQ denotes the pair of OpenVidHD and DynamicCrafter. The results validate the effectiveness of the proposed quality coverage.

Expansion		Real Testing Data						Overall Mean
LQ	UHQ	Youku-mPLUG	MSR-VTT	Vript	HD-VG-130M	RealVSR	UltraVideo	
-	-	89.02	93.03	89.12	84.51	89.39	81.58	87.78
✓	-	94.15	97.30	95.07	91.08	90.39	87.35	92.56
-	✓	91.59	92.25	92.52	89.48	90.12	89.23	90.86
✓	✓	93.72	95.21	95.53	91.86	92.48	91.54	93.39

Table 3. Evaluation of data efficiency and scalability. Intra-dataset results are reported in *italics*. Biased++ is composed of three LQ real sources and three HQ generators. Mean* denotes the average AUC on unseen datasets. K400, Youku, and HD-VG refer to Kinetics-400, Youku-mPLUG, and HD-VG-130M, respectively.

Variant	K400	Youku	MSR-VTT	InternVid-AES	Vript	HD-VG	OpenVidHD	RealVSR	UltraVideo	Overall Mean	Mean*
Biased	<i>95.54</i>	91.54	95.36	74.90	86.62	81.40	76.28	74.24	61.03	81.88	75.82
Biased++	<i>98.01</i>	98.36	99.53	86.26	94.89	92.93	82.90	80.79	70.90	89.40	84.88
Aligned	84.32	89.02	93.03	93.29	89.12	84.51	90.85	89.39	81.58	88.35	86.15
Expanded	<i>95.11</i>	93.72	95.21	95.67	95.53	91.86	96.97	92.48	91.54	94.23	92.85

(LQ), Vript and HD-VG-130M (HQ), and RealVSR and UltraVideo (UHQ). For performance evaluation, we pair each real dataset with the nine generative models from GenVideo testset to calculate the mean AUC. Finally, we report the Overall Mean across all six real datasets to provide a holistic performance summary.

Validation across Backbones. To validate the effectiveness and universality of our quality-aware data alignment and expansion, we selected seven distinct detector backbones for comparative analysis. These include two image-level models (DINOv2 and DINOv3) and three video-level models (TimeSformer, VideoMAEv2, and DeMamba). For each backbone, we conduct experiments under three distinct training configurations:

- **Biased:** The original Kinetics-400 and Pika pair, representing the baseline setting with a marked quality mismatch.
- **Aligned:** The InternVid-AES and Pika pair, constructed via our proposed quality-aware alignment.
- **Expanded:** The comprehensive training set obtained by applying quality-aware data expansion on top of the aligned baseline.

Table 1 details the quantitative performance of seven detector backbones across three distinct training settings. Evidently, the *Aligned* setting delivers an average gain of 3.18% over the *Biased* baseline, with clear improvements observed across most backbones. This dominant upward trend affirms the broad efficacy of our quality-aware data alignment. Building upon this foundation, the *Expanded* configuration provides an additional boost, elevating the Overall Mean by an average of 6.06% and proving the effectiveness of our data expansion strategy. Taken together, these results

compellingly demonstrate that our proposed paradigm significantly bolsters the generalization capabilities of existing detectors across diverse architectures.

4.4. Ablation Study on Quality-aware Data Expansion

Consistent with Section 3.2, we employ VideoMAEv2 as the representative backbone to conduct an ablation study on our quality-aware data expansion strategy. We compare our full expansion method against three alternative configurations: the original InternVid-AES and Pika baseline, a variant with only LQ data expansion (Kinetics-400 and OpenSora), and a variant with only UHQ data expansion (OpenVidHD and DynamicCrafter).

The experimental results are presented in Table 2. It is observed that the incorporation of either LQ or UHQ data significantly enhances model generalization. Notably, the model achieves optimal performance when both data sources are utilized simultaneously, validating the effectiveness of our quality-aware data expansion. Furthermore, we observe that the inclusion of LQ data exerts the most substantial impact. We attribute this to the fact that the LQ interval is the most sparse region within the original InternVid-AES and Pika distribution, and the introduction of LQ data effectively compensates for this deficiency.

4.5. Analysis of Data Efficiency and Scalability

To evaluate data efficiency and scalability, we constructed Biased++—a $3\times$ larger training set mixing three LQ real sources (Kinetics-400, Youku-mPLUG, MSR-VTT) and three HQ generators (Pika, SEINE, SD). We also employ VideoMAEv2 as the representative backbone.

Results in Table 3 show that our quality-matching strategy consistently outperforms baselines at equivalent scales

Table 4. Generalization results on GenVidBench. The best results are marked in **bold**.

Model	Biased	Aligned	Expanded
DINOv2	84.32	89.09	91.17
DeMamba	83.83	85.22	85.86
VideoMAEv2	79.49	79.59	85.60

(Aligned vs. Biased and Expanded vs. Biased++). Notably, even with a $3\times$ data scale disadvantage, Aligned achieves generalization performance comparable to Biased++. These results demonstrate the superior data efficiency and scalability of our approach.

4.6. Verification of Protocol Effectiveness

We also conduct additional cross-dataset evaluations on GenVidBench to verify the effectiveness of our revised protocol. As shown in Table 4, the *Expanded* training setting consistently achieves the best performance across different backbones. Furthermore, when compared with the original GenVideo protocol (which assesses generalization via Youku-mPLUG and MSR-VTT), the performance trends under our protocol exhibit stronger agreement with these GenVidBench cross-dataset results, as evidenced by Table 4 and Table 1. This results demonstrate that our revised protocol not only produces more generalizable models, but also offers a more reliable reflection of real-world deployment performance.

5. Discussion

5.1. Rethinking Real Data Quality in Current Benchmarks

Guided by the quality analysis established in previous sections, we scrutinize the real data quality distributions of several mainstream AIGV detection benchmarks: GenVideo, GenVidBench, GenBuster-200K, and GenBuster++.

GenVideo employs Kinetics-400, Youku-mPLUG, and MSR-VTT for training or evaluation. As illustrated in Fig. 3, these three datasets exhibit a consistent quality profile, being universally low (LQ) and closely clustered in terms of distribution.

GenVidBench utilizes Vript for training and HD-VG-130M for testing. Our analysis indicates that both datasets share a comparable quality level, predominantly falling within the High Quality (HQ) range.

GenBuster-200K and GenBuster++ rely on OpenVid-1M and OpenVidHD, respectively, with the latter being a high-quality subset of the former. Corroborated by both the original OpenVid quality reports and our previous assessment, the real data in these benchmarks is characterized by Ultra High Quality (UHQ).

In summary, the real data utilized in current benchmarks is predominantly confined to narrow, localized regions of the quality spectrum. Although recent benchmarks (e.g., GenVidBench, GenBuster-200K, and GenBuster++) have begun to emphasize the quality and variety of real videos, they exhibit a clear bias towards high-fidelity data, overlooking the prevalence of low-quality videos in real-world scenarios. Regarding the design of training and evaluation protocols, although certain benchmarks (e.g., GenVideo and GenVidBench) employ distinct real datasets for training and testing to probe model generalization, the selected datasets exhibit remarkably similar quality profiles. Consequently, this quality homogeneity prevents an objective assessment of the model’s usability in complex real-world scenarios. Our proposed revised protocol effectively bridges this gap, providing a complete picture of model performance across the full quality spectrum.

5.2. Remarks on Quality Dynamics

It is foreseeable that with rapid technological advancements, the quality of both real-world videos and generative models will continue to escalate. Consequently, their respective quality distributions are not static, posing increasingly higher demands on detection systems. We hope that the regularities uncovered and the methodology proposed in this work will offer valuable insights for addressing these evolving challenges.

Additionally, we acknowledge that our classification of quality tiers is coarse-grained and empirical, as real-world datasets inevitably contain a small fraction of samples distributed outside their primary quality intervals. Moreover, while DOVER serves as a proxy for human perception and is subject to inherent prediction errors, our reliance on statistical results derived from large-scale data effectively mitigates sample-level noise, ensuring the reliability of our findings.

6. Conclusion

This paper identifies and addresses a critical shortcut learning issue in AI-generated video detection, where models rely on video quality discrepancies rather than actual forgery artifacts. To address this, we proposed a quality distribution alignment and data expansion strategy that significantly improves model generalization, a capability validated by extensive experiments. Furthermore, we exposed the design deficiencies in current benchmarks and introduced a revised evaluation protocol to more accurately measure a detector’s real-world performance. However, being relatively passive compared to active augmentation techniques, this leaves exploring proactive methods an open question. Ultimately, we hope this work provides valuable empirical references and theoretical guidance for the AIGV community, facilitating the development of more reliable detection methods.

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Impact Statement

This paper investigates AI-generated video detection, which has emerged as a critical topic for digital content governance as synthetic videos become increasingly perceptually indistinguishable from real footage. The findings in this paper—particularly regarding the impact of data quality distributions—reveal the limitations of current benchmarks and are expected to promote the development of more generalizable models capable of handling diverse video qualities. A potential negative impact lies in the facilitation of adversarial evasion attacks against existing detectors. For instance, malicious actors might intentionally degrade video resolution or compression quality to impair the performance of detectors trained on high-quality data. However, we believe that highlighting this vulnerability is essential, as it motivates future research on quality-resilient and robust detectors for AI-generated videos in complex real-world scenarios.

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A. Dataset Details

A.1. Real Datasets

Here, we present supplementary details, specifications, and sample visualizations for the real video datasets utilized in our experiments.

Kinetics-400 (Kay et al., 2017) is a large-scale human action video dataset comprising 400 action classes, originally released in 2017.



Figure 7. Sample visualization of Kinetics-400.

MSR-VTT (Xu et al., 2016) is a large-scale benchmark dataset for video description, containing 10,000 web video clips. It encompasses 20 representative categories, including music, people, gaming, sports, news, education, TV shows, movies, animation, and cooking. It is originally released in 2016.



Figure 8. Sample visualization of MSR-VTT.

Youku-mPLUG (Xu et al., 2023) is a high-quality Chinese video-language dataset from Youku, featuring 10 million pre-training pairs and a 300,000-sample human-annotated benchmark, originally released in 2023.



Figure 9. Sample visualization of Youku-mPLUG.

InternVid-AES (Wang et al., 2023c) is a subset of InternVid consisting of ~ 5.6 million video-text pairs. It focuses on high visual quality, selected via an aesthetic scoring model to represent videos with significant aesthetic value, originally released in 2023.



Figure 10. Sample visualization of InternVid-AES.

Vript (Yang et al., 2024) is a large-scale, high-definition video description dataset sourced from YouTube. It encompasses over 12,000 video scenes and covers more than 1,300 distinct action categories, originally released in 2024.



Figure 11. Sample visualization of Vript.

HD-VG-130M (Wang et al., 2025a) is a large-scale, open-source dataset comprising 130 million text-video pairs. It is distinguished by its high visual quality, featuring content that is exclusively high-definition (720p), widescreen, and watermark-free, originally released in 2023.



Figure 12. Sample visualization of HD-VG-130M.

OpenVidHD (Nan et al., 2024) is a high-definition subset comprising 433,000 video clips with resolutions reaching the 1080p level. It was carefully curated from the larger OpenVid-1M dataset, which consists of 1 million videos at 720p resolution, originally released in 2024.



Figure 13. Sample visualization of OpenVidHD.

RealVSR (Yang et al., 2021) is a real-world video super-resolution dataset captured using the dual-camera system of an iPhone 11 Pro Max. It comprises 500 pairs of frame-aligned Low-Resolution (LR) and High-Resolution (HR) sequences, recorded across diverse lighting conditions and scenes, originally released in 2021.

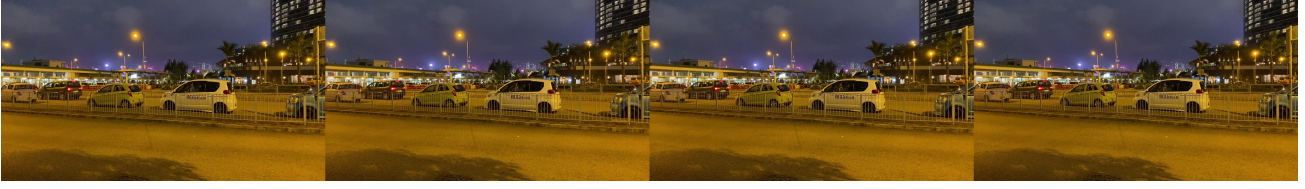


Figure 14. Sample visualization of RealVSR.

UltraVideo (Xue et al., 2025) is an open-source video dataset dedicated to native 4K/8K Ultra-High Definition (UHD) resolutions. It comprises approximately 42,000 short clips (3–10 seconds) and 17,000 long videos (≥ 10 seconds), spanning 108 distinct thematic categories, originally released in 2025.



Figure 15. Sample visualization of UltraVideo.

A.2. Video Generative Models

Table 5. Details of video generative models in the GenVideo dataset.

Generative Model	Split	Type	Resolution	FPS
ZeroScope (ZeroScope)	Train	T2V	1024×576	8
I2VGen-XL (Zhang et al., 2023)		I2V	1280×720	8
SVD (Blattmann et al., 2023)		I2V	1024×576	8
VideoCrafter (Chen et al., 2024b)		T2V	1024×576	8
Pika (Pika, 2025)		T2V&I2V	1088×640	24
DynamiCrafter (Xing et al., 2024)		I2V	1024×576	8
SD (Zhang et al., 2024)		T2V&I2V	512-1024	8
SEINE (Chen et al., 2023b)		I2V	1024×576	8
Latte (Ma et al., 2025)		T2V	512×512	8
OpenSora (Zheng et al., 2024)		T2V	512×512	8
ModelScope (Wang et al., 2023a)	Test	T2V	256×256	8
MorphStudio (MorphStudio, 2025)		T2V	1280×720	8
MoonValley (Moonvalley, 2025)		T2V	1024×576	16
HotShot (Mullan et al., 2023)		T2V	672×384	8
Show_1 (Zhang et al., 2025a)		T2V	576×320	8
Gen2 (Esser et al., 2023)		I2V&T2V	896×512	24
Crafter (Chen et al., 2023a)		T2V	256×256	8
Lavie (Wang et al., 2025b)		T2V	1280×2048	8
Sora (OpenAI, 2025)		T2V	-	-
WildScape (Wei et al., 2024; Feng et al., 2023; Xu et al., 2024)		T2V&I2V	512-1024	8-16

Tab. 5 provides detailed specifications of the generative models included in the GenVideo dataset. During our experiments, we made specific exclusions to ensure data consistency:

HotShot: We excluded this model from the testing phase because a significant portion of its generated videos contain fewer than 16 frames. This insufficient temporal resolution fails to meet the minimum input requirements of the VideoMAEv2 backbone.

ZeroScope: We opted not to utilize ZeroScope for training data expansion due to a naming anomaly observed in the GenVideo dataset. Specifically, the files purportedly belonging to ZeroScope were erroneously labeled with the 'VideoCrafter' prefix. Given the ambiguity regarding whether this was a nomenclature error or a file misplacement, we excluded these samples to maintain data integrity.

B. More Experimental Details and Results

B.1. Experimental Details

Data Preprocessing. For data preprocessing, we strictly follow the official guidelines provided by the backbone authors. It is worth noting that for X-CLIP (Ni et al., 2022), we reverted to its native preprocessing implementation, in contrast to the modified setting adopted in DeMamba.

Reproducibility. To ensure reproducibility, we fixed the global random seed to 42 for all experiments. However, it is worth noting that the preprocessing pipeline for DINOv2 involves the `upsample_bicubic2d_backward_out_cuda` operation. As this specific CUDA kernel lacks a deterministic implementation, minor variations in the DINOv2's experimental results may occur.

B.2. DeMamba Trained on Different Real Datasets

We further conducted experiments using DeMamba, training it separately on three datasets representing distinct quality levels. As illustrated in Fig. 16, the results indicate that DeMamba exhibits a trend consistent with that of VideoMAEv2.

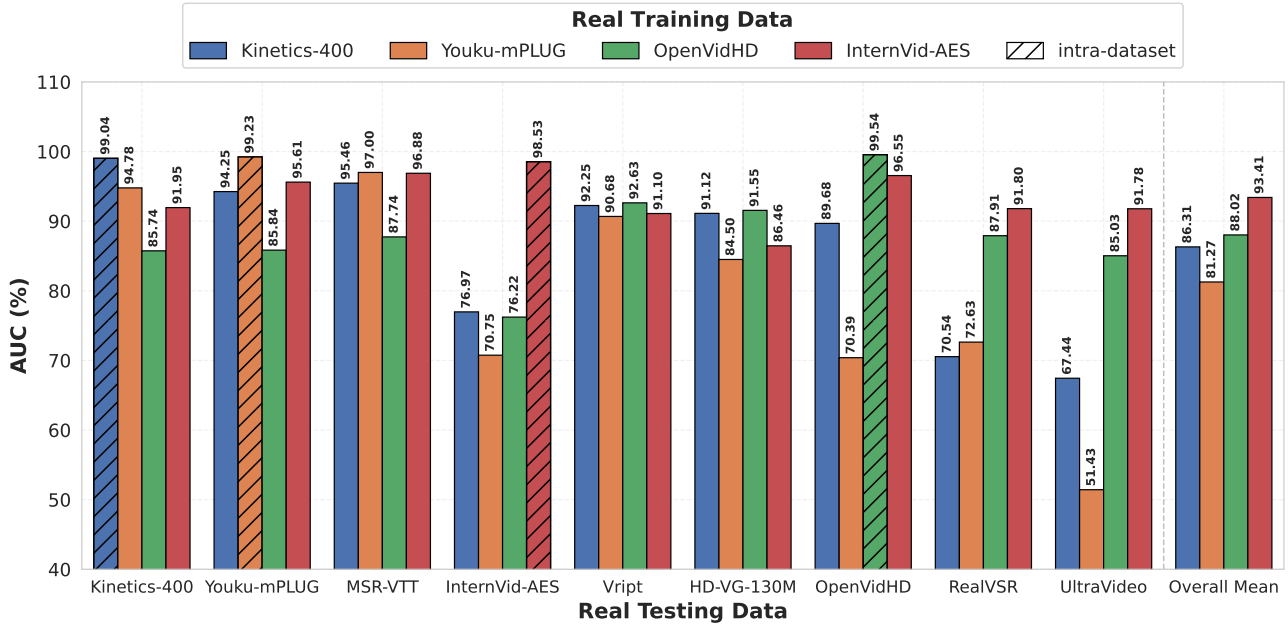


Figure 16. Cross-dataset generalization performance of DeMamba trained on different real datasets. The hatched bars represent intra-dataset evaluation results.

B.3. VideoMAEv2 Trained on Vript

We further trained VideoMAEv2 on Vript, with generalization results presented in Fig. 17. While the overall trend mirrors that of InternVid-AES, Vript surpasses it on LQ real datasets but lags behind on UHQ ones. This observation aligns with the quality statistics in Fig. 3, which indicate that Vript possesses a slightly lower quality profile than InternVid-AES, thereby enabling the trained model to generalize better to lower-quality samples.

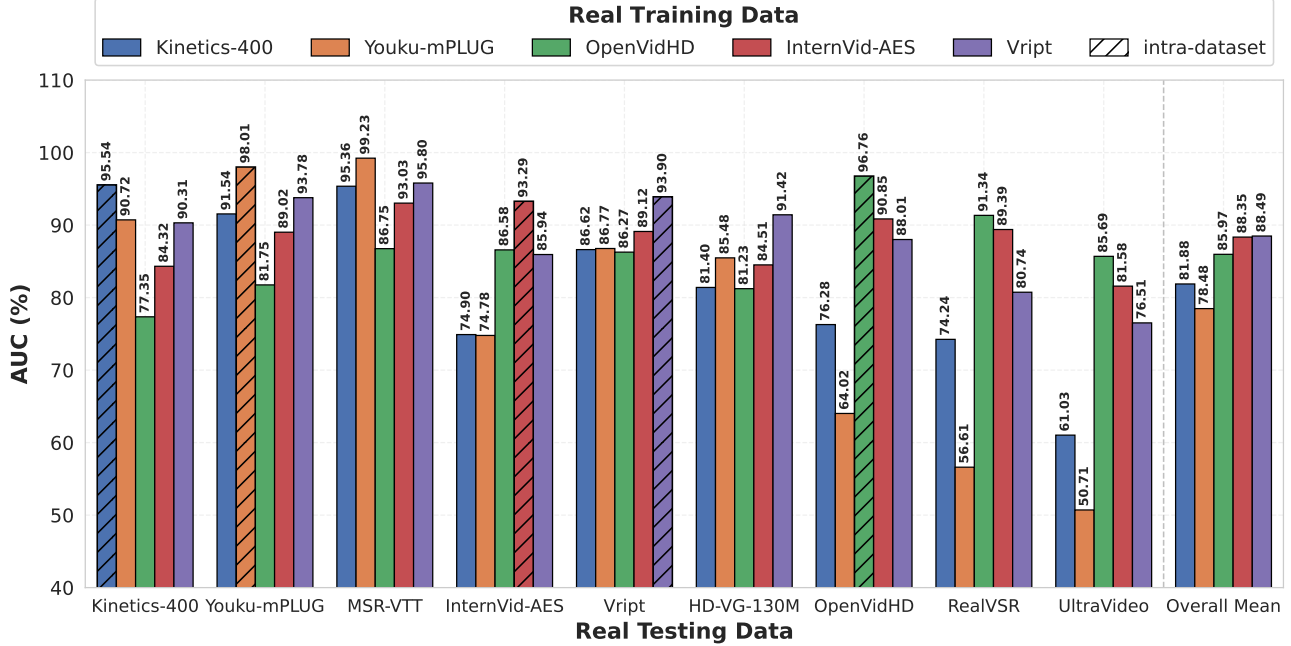


Figure 17. Cross-dataset generalization performance of VideoMAEv2 trained on Vript.

B.4. Comparison between DDA and Our Approach

To further clarify the distinction between our work and DDA (Chen et al., 2025), we conducted a direct comparative experiment using DINOv2 as the backbone. We evaluated DDA’s active alignment against our selection-based approach on both Kinetics-400 (Biased) and InternVid-AES (Aligned) distributions.

The results in Table 6 demonstrate that DDA’s performance on video data is suboptimal compared to our approach. We attribute this to the following fundamental differences:

- **Spatial vs. Temporal Artifacts:** DDA is designed for image-level alignment, utilizing 2D frequency-domain processing and image-based VAE reconstruction. However, modern video generators increasingly adopt Video-level VAEs and temporal modules. DDA’s pixel-level manipulation lacks the capacity to address temporal artifacts and may even introduce jittering or unnatural smoothness, compromising the detector’s ability to learn robust representations.
- **Authenticity vs. Manipulation:** While DDA “actively” alters pixels to force alignment—which may inadvertently introduce specific reconstruction patterns—our method preserves the authenticity of both real and generated content through intelligent selection. This ensures the detector learns intrinsic forgery features rather than artifacts introduced by the alignment process itself.

The substantial performance gap proves that “quality-matching” is not merely an incremental application of DDA’s principles. Instead, it offers a distinct, more reliable path for video forgery detection by maintaining content integrity while effectively mitigating the quality shortcut.

Table 6. Performance comparison between DDA and our approach (Backbone: DINOv2).

Variant	Youku-mPLUG	MSR-VTT	Vript	HD-VG-130M	RealVSR	UltraVideo	Overall Mean
DDA (Kinetics-400)	88.63	77.32	75.80	73.43	48.14	36.57	66.65
DDA (InternVid-AES)	61.79	27.63	75.62	68.96	77.68	67.94	63.27
Biased (Kinetics-400+Pika)	94.97	93.19	89.85	87.91	74.98	63.53	84.07
Aligned (InternVid-AES+Pika)	94.56	91.88	89.86	85.79	93.29	83.46	89.81